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CABLE CONNECTOR ASSEMBLY HAVING CONNECTOR IDENTIFICATION MODULE

Abstract

A connector assembly for mating with a mating connector assembly includes a connector holder that has a frame defining a connector chamber configured to be mounted to a panel. The connector assembly includes a cable connector received in the connector chamber and extending forward from the front of the frame to pass through an opening in the panel for mating with the mating connector. The connector assembly includes a connector identification module held by the frame including a mating end configured to be mated with a mating identification module of the mating connector assembly. The connector identification module includes an identification device having unique identification data and operably coupled to the identification contacts to transmit the unique identification data to the mating identification module.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application claims benefit to U.S. Provisional Application No. 63/563,670, filed 11 Mar. 2024, titled “CABLE CONNECTOR ASSEMBLY HAVING CONNECTOR IDENTIFICATION MODULE”, the subject matter of which is herein incorporated by reference in its entirety.

BACKGROUND OF THE INVENTION

[0002] The subject matter herein relates generally to communication systems.

[0003] Cable assemblies are used to electrically connect various components within a system, such as a server or network system. The cable assemblies extend between the components, such as network, compute, storage, memory, or other components. The components may be arranged in shelves or trays of a network rack. Some systems may include components such as cards, such as line cards, daughter cards, mother boards, and the like. Each cable assembly includes multiple cables arranged in a bundle or array. Each cable forms a data channel. As the density of cable assemblies in the network system increase, it is desirable to know the locations of the components within the system, the manufacturing information of components and other device readable information, such as for service, repair or replacement.

[0004] A need remains for a method and device for identifying components and locations of components within a network system.

BRIEF DESCRIPTION OF THE INVENTION

[0005] In one embodiment, a connector assembly for mating with a mating connector assembly is provided. The connector assembly includes a connector holder that has a frame defining a connector chamber. The frame extends between a front and a rear of the connector holder. The frame includes an upper frame member, a lower frame member, and first and second side frame members extend between the upper and lower frame members to define the connector chamber. The connector holder configured to be mounted to a panel. The connector assembly includes a cable connector received in the connector chamber. The cable connector includes a connector housing holding contact assemblies. Each contact assembly includes a signal contact and a cable terminated to the signal contact. The signal contact configured to be mated with a mating signal contact of a mating connector of the mating connector assembly. The cable connector has a mating end that extends forward from the front of the frame and configured to pass through an opening in the panel for mating with the mating connector in a mating direction. The connector assembly includes a connector identification module is held by the frame and includes a mating end configured to be mated with a mating identification module of the mating connector assembly. The connector identification module includes identification contacts and an identification device. The identification contacts arranged at the mating end and configured to be mated with the mating identification module. The identification device has unique identification data stored on the identification device. The identification device is operably coupled to the identification contacts to transmit the unique identification data to the mating identification module.

[0006] In another embodiment, a connector assembly for mating with a mating connector assembly is provided. The connector assembly includes a connector holder that has a frame defining a connector chamber. The frame extends between a front and a rear of the connector holder. The frame includes an upper frame member, a lower frame member, and first and second side frame members extend between the upper and lower frame members to define the connector chamber. The connector holder configured to be mounted to a panel. The connector assembly includes a

cable connector received in the connector chamber. The cable connector includes a connector housing holding contact assemblies. Each contact assembly includes a signal contact and a cable terminated to the signal contact. The signal contact configured to be mated with a mating signal contact of a mating connector of the mating connector assembly. The cable connector has a mating end that extends forward from the front of the frame and configured to pass through an opening in the panel for mating with the mating connector in a mating direction. The connector assembly includes a connector identification module held by the frame and includes a mating end configured to be mated with a mating identification module of the mating connector assembly. The connector identification module includes a printed circuit board that has a card edge at the mating end. The connector identification module includes identification contacts arranged at the card edge. The connector identification module includes an identification device mounted to the printed circuit board. The identification device has unique identification data stored on the identification device. The identification device is operably coupled to the identification contacts, wherein the card edge is configured to be plugged into a card slot of the mating identification module when the connector assembly is mated with the mating connector assembly to transmit the unique identification data to the mating identification module.

[0007] In a further embodiment, a communication system is provided and includes a cartridge that has panels forming a chamber. The panels include a front panel. The cartridge includes ports in the front panel at a front of the cartridge. The communication system includes connector assemblies received in the ports for mating with mating connector assemblies. Each connector assembly includes a connector holder, a cable connector held by the connector holder, and a connector identification module for identifying the cable connector within the communication system. Each connector holder has a frame defining a connector chamber. The frame extends between a front and a rear of the connector holder. The frame includes an upper frame member, a lower frame member, and first and second side frame members extend between the upper and lower frame members to define the connector chamber. The connector holder is mounted to the cartridge at the corresponding port. Each cable connector is received in the corresponding connector chamber. Each cable connector includes a connector housing holding contact assemblies. Each contact assembly includes a signal contact and a cable terminated to the signal contact. The cable extends from the cable connector into the chamber of the cartridge. The signal contact configured to be mated with a mating signal contact of a mating connector of the corresponding mating connector assembly. The cable connector has a mating end that extends forward from the front of the frame and is configured to pass through the corresponding port in the front panel for mating with the mating connector in a mating direction. Each connector identification module is held by the frame proximate to the cable connector. The connector identification module includes a mating end configured to be mated with a mating identification module of the corresponding mating connector assembly. The connector identification module includes identification contacts and an identification device. The identification contacts arranged at the mating end and configured to be mated with the mating identification module. The identification device has unique identification data stored on the identification device. The identification device is operably coupled to the identification contacts to transmit the unique identification data to the mating identification module.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. 1 is a front view of a communication system in accordance with an exemplary embodiment.

[0009] FIG. 2 is a front view of a portion of the communication system in accordance with an exemplary embodiment.

[0010] FIG. 3 illustrates a portion of the communication system showing two of the cartridges in accordance with an exemplary embodiment.

[0011] FIG. 4 is a perspective view of a portion of the communication system showing one of the mating connector assemblies poised for coupling to the corresponding connector assembly in accordance with an exemplary embodiment.

[0012] FIG. 5 is a front perspective view of a portion of the communication system showing one of the connector assemblies in accordance with an exemplary embodiment coupled to the front panel of the cartridge.

[0013] FIG. 6 is a front perspective, exploded view of the connector assembly in accordance with an exemplary embodiment.

[0014] FIG. 7 is a front perspective view of one of the contact modules in accordance with an exemplary embodiment.

[0015] FIG. 8 is an enlarged view of the mating end of the contact module in accordance with an exemplary embodiment.

[0016] FIG. 9 is a cross sectional view of a portion of the communication system showing the mating connector assembly mated with the connector assembly showing the connector identification module in accordance with an exemplary embodiment.

[0017] FIG. 10 is a cross sectional view of a portion of the communication system showing the mating connector assembly mated with the connector assembly showing the connector identification module in accordance with an exemplary embodiment.

[0018] FIG. 11 is a perspective view of the connector identification module in accordance with an exemplary embodiment.

[0019] FIG. 12 is a top view of a portion of the communication system in accordance with an exemplary embodiment showing the mating connector assembly mated with the connector assembly showing the connector identification module shown in FIG. 11.

[0020] FIG. 13 is a side view of a portion of the communication system in accordance with an exemplary embodiment showing the mating connector assembly mated with the connector assembly showing the connector identification module shown in FIG. 11.

[0021] FIG. 14 is a rear perspective view of a portion of the communication system in accordance with an exemplary embodiment showing the mating connector assembly mated with the connector assembly showing the connector identification module shown in FIG. 11.

DETAILED DESCRIPTION OF THE INVENTION

[0022] FIG. 1 is a front view of a communication system **100** in accordance with an exemplary embodiment. FIG. 2 is a front view of a portion of the communication system **100** in accordance with an exemplary embodiment. In an exemplary embodiment, the communication system **100** includes a chassis or rack **50** having a rack frame **52** holding electrical components **60**. The rack **50** may be a server rack. The electrical components **60** may include fixed modules **62** (FIG. 2) and/or removable modules **64** (FIG. 1). The removable modules **64** may be mated to the fixed modules **62**. For example, the removable modules **64** may be plugged into the rack frame **52** and removable from the rack frame **52**. In various embodiments, the electrical components **60** may include one or more patch panels, switches, servers, routers, firewalls, and the like. The rack frame **52** may hold a power supply, cooling components, and the like.

[0023] In an exemplary embodiment, the communication system **100** includes at least one connector assembly **102** (FIG. 2) and at least one mating connector assembly **70** (FIG. 4) configured to be mated with the corresponding at least one connector assembly **102**. The connector assembly **102** may be part of one of the electrical components **60**, such as one of the fixed modules **62**. The mating connector assembly **70** may be part of one of the electrical components **60**, such as one of the removable modules **64**. In various embodiments, each mating connector assembly **70** may include one or more mating electrical connectors mounted to a circuit board, such as a backplane, a daughter card, a network switch, and the like.

[0024] In an exemplary embodiment, the communication system **100** includes a connector identification system **400** to identify the various connector assemblies **102** and/or mating connector assemblies **70**. The connector identification system **400** may be used to identify physical locations of the connector assemblies **102** within the communication system **100**, such as to identify which connector in a cable harness is electrically connected to which electrical component **60** for system management. In an exemplary embodiment, the connector identification system **400** may use identification devices, such as EEPROMs (Electrically Erasable and Programmable Read Only Memory), to hold information that allows identification of the various connector assemblies **102**. The identification devices may be incorporated with each of the connector assemblies **102**. For example, the identification devices may be incorporated into the connector assembly **102**, such as housed within the connector assembly and having a mating interface configured to mate with the mating connector assembly **70**. The identification devices may have unique identification information programmed into the identification devices, such as a part number, a program name, a slot ID number, data about the associated connector assembly **102**, and the like. In an exemplary embodiment, the identification devices also include supporting componentry like Resistors, Capacitors, and other supporting Integrated Circuits. In an exemplary embodiment, the identification devices are modular and may be incorporated into different types of connector assemblies **102**.

[0025] In an exemplary embodiment, the communication system **100** includes a cabinet or cartridge **110** that holds the connector assemblies **102**. The connector assemblies **102** may be cable connector assemblies having cables extending from corresponding cable connectors. The cartridge **110** may form part of the fixed modules **62**. Optionally, multiple cartridges **110** may be provided (for example, left side and right side). The cartridge **110** may be coupled to the rack frame **52**, such as at the rear of the rack frame **52**. The cartridge **110** forms an enclosed space or chamber **111** for the connector assemblies **102** and the cables of the connector assemblies **102**. The cartridge **110** is used for cable management. For example, the cables may be part of cable harnesses routed within the enclosed chamber **111** formed by the cartridge **110** electrically connecting the various connector assemblies **102**. The cartridge **110** is used to support or position the connector assemblies **102** for mating with the mating connector assemblies **70**.

[0026] FIG. 3 illustrates a portion of the communication system **100** showing two of the cartridges **110** in accordance with an exemplary embodiment. Each cartridge **110** holds a plurality of the connector assemblies **102**, such as in one or more rows and one or more columns. The connector assemblies **102** are provided at the front of the cartridge **110** for interfacing with the mating connector assemblies **70** (shown in FIG. 4).

[0027] In an exemplary embodiment, the cartridge **110** is formed from a plurality of panels **112**, such as sheet metal panels. The cartridge **110** includes a front panel **114**, side panels **116**, and a rear panel **118**. The panels **112** may additionally include an upper panel and/or a lower panel. The connector assemblies **102** are provided at the front panel **114** for mating with the mating connector assemblies **70**. In an exemplary embodiment, the connector assemblies **102** are arranged in a column along the front panel **114**. Optionally, the connector assemblies **102** may be provided in multiple columns. The mating ends of the connector assemblies **102** are external to the cartridge **110**. For example, the connector assemblies **102** pass through ports or openings **120** in the front panel **114**. The connector assemblies **102** are coupled to the front panel **114** adjacent the openings **120**. In an exemplary embodiment, the connector assemblies **102** may float or move relative to the front panel **114** to align with and mate to the mating connector assemblies **70**. In an exemplary embodiment, the connector assemblies **102** include mating guides, such as guide posts **122**, to guide mating of the mating electrical connectors **72** with the connector assemblies **102**. The guide posts **122** extend forward of the front panel **114**.

[0028] FIG. 4 is a perspective view of a portion of the communication system **100** showing one of the mating connector assemblies **70** coupled to the corresponding connector assembly **102**. The

connector assembly **102** is coupled to the cartridge **110**, such as to the front panel **114**. In the illustrated embodiment, the mating connector assembly **70** is a network component. For example, the mating connector assembly **70** may be a compute node, a memory module, a network switch, a storage device, a network accelerator, or another type of communication component. The mating connector assembly **70** may be a backplane component or a daughtercard component.

[0029] In an exemplary embodiment, the connector assembly **102** includes a connector identification module **402**, forming part of the connector identification system **400**. The connector identification module **402** is associated with the cable connector(s) of the connector assembly **102**. The connector identification module **402** is used to identify the connector assembly **102** within the communication system **100**. The connector identification module **402** is integrated with the other components of the connector assembly **102** for mating with the corresponding mating connector assembly **70**.

[0030] In an exemplary embodiment, the connector identification module **402** includes an identification device **410** holding unique identification data. The identification device **410** is able to uniquely identify the connector assembly **102**. For example, the identification device **410** is able to identify the cable connector from a plurality of cable connectors, such as within the cartridge **110**. The identification device **410** may identify the location of the connector assembly **102** or other identifying information about the cable connector(s) or other components of the connector assembly **102**. In an exemplary embodiment, the identification device **410** is an EEPROM device that is programmable and configured to store unique identification data.

[0031] In an exemplary embodiment, the mating connector assembly **70** includes a mating electrical connector **72** mounted to a circuit board **74**. The mating connector assembly **70** may be mounted to a front edge of the circuit board **74**. The mating electrical connector **72** includes a connector housing **76** holding signal contacts (not shown) and ground shields shielding the signal contacts. The signal contacts may be arranged in pairs. The signal contacts may be spring beams in various embodiments. The signal contacts may be pins, sockets, or other types of contacts in alternative embodiments. The signal contacts are terminated to the circuit board **74**.

[0032] In an exemplary embodiment, the mating connector assembly **70** includes a mating identification module **80** mounted to the circuit board **74**. The mating identification module **80** is configured to be mated with the connector identification module **402** of the connector assembly **102**. The unique identifying data is configured to be transmitted from the connector identification module **402** to the mating identification module **80** when the mating connector assembly **70** is mated with the connector assembly **102**. In an exemplary embodiment, the mating identification module **80** includes an identification connector, such as a card edge connector **82**, mounted to the circuit board **74**.

[0033] The card edge connector **82** includes a card slot **84** configured to receive a portion of the connector identification module **80**. In the illustrated embodiment, the card slot **84** is oriented vertically. In an exemplary embodiment, the connector identification module **402** is an orthogonal connector. For example, the mating interface of the connector identification module **402** is oriented perpendicular to the circuit board **74**. The mating identification module **80** may include additional components, such as a microprocessor, an EEPROM chip, or other type of component, mounted to the circuit board **74** operably coupled to the card edge connector **82** for controlling communication between the connector assembly **102** and the mating connector assembly **70**. The mating identification module **80** may include a communication module, such as a wireless communication device, such as a transmitter or other type of wireless communication device, to transmit the data from the connector identification module **402** to a remote device, such as a central processing unit for the communication system **100**.

[0034] The connector assembly **102** is received in the cartridge **110**. The connector assembly **102** is provided at the front panel **114** for mating with the mating connector assembly **70** along a mating axis. In the illustrated embodiment, the mating axis extends along the Z-axis. The mating end of the

cable connector **130** is configured to pass through an opening or port in the front panel **114** for mating with the mating electrical connector **72**. A mating end of the connector identification module **402** may pass through an opening in the front panel **114** for mating with the mating identification module **80**. The connector assemblies **102**, **70** are configured to transmit and/or receive data through an interface. In an exemplary embodiment, the identification devices of the connector identification system **400** are integrated with the connector assemblies **102** to mate with the mating connector assemblies **70**. For example, the connector identification module **402** may directly mate with the mating identification module **80** when the mating connector assembly **70** is mated to the connector assembly **102**. The connector assemblies **102** may include receptacle connectors and the mating connector assemblies **70** may include plug connectors. In an exemplary embodiment, the guide post **122** is configured to be mated with a guide module (not shown) of the mating connector assembly **70** to guide mating of the mating connector assembly **70** with the connector assembly **102**. The guide module may be mounted to the circuit board **74** and/or the mating electrical connector **72**. The guide post **122** and the guide module may guide mating of the connector identification module **402** and the mating identification module **80**. In an exemplary embodiment, the guide module includes an opening that receives the guide post **122**. The mating guides may be used to align the connectors in one or more lateral directions transverse to the mating axis, such as along the X-axis and/or the Y-axis. In various embodiments, the mating guides may provide both horizontal alignment (X-axis) and vertical alignment (Y-axis).

[0035] FIG. 5 is a front perspective view of a portion of the communication system **100** showing one of the connector assemblies **102** in accordance with an exemplary embodiment coupled to the front panel **114** of the cartridge **110**. FIG. 6 is a front perspective, exploded view of the connector assembly **102** in accordance with an exemplary embodiment. The connector assembly **102** includes one or more cable connectors **130**, a connector holder **200** holding the cable connector(s) **130**, a mounting assembly **300** for mounting the connector holder **200** and the cable connector(s) **130** to the front panel **114** of the cartridge **110**, and a connector identification module **402** associated with the cable connector(s) **130**. The connector identification module **402** is part of the connector identification system **400** used to identify the connector assembly **102** within the system. The connector identification module **402** is integrated with the cable connector **130** in the connector holder **200**. The connector identification module **402** is configured to be mated to the mating connector assembly **70** with the cable connector **130**.

[0036] In an exemplary embodiment, the connector identification module **402** includes an identification device **410** holding unique identification data. The identification device **410** is able to uniquely identify the cable connector **130**. For example, the identification device **410** is able to identify the cable connector **130** from a plurality of cable connectors, such as within the cartridge **110**. The identification device **410** may identify the location of the cable connector **130** or other identifying information about the cable connector **130**. In an exemplary embodiment, the identification device **410** is an EEPROM device that is programmable and configured to store unique identification data.

[0037] In the illustrated embodiment, the connector assembly **102** includes two cable connectors **130** arranged side-by-side. However, greater or fewer cable connectors **130** may be provided in alternative embodiments. The cable connectors **130** may be vertically stacked rather than horizontally stacked in alternative embodiments. The mating ends of the cable connectors **130** pass through the opening **120** in the front panel **114** for mating with the mating electrical connector **72** (FIG. 4).

[0038] The cable connector **130** includes a connector housing **132** holding a plurality of communication contact modules **160** configured to transmit data, such as high speed data signals. The contact modules **160** include contact assemblies **140** configured to be mated with the mating connector assembly **104**. The contact assemblies **140** are terminated to ends of cables **150** extending from the cable connector **130**. For example, the cables **150** may extend from the rear

ends of the contact modules **160**. The contact assemblies **140** are arranged together within the contact modules **160**. For example, the contact modules **160** hold the contact assemblies **140** in columns. A plurality of the contact modules **160** are received in the connector housing **132**, such as being arranged in a contact module stack.

[0039] The connector housing **132** includes a cavity **134** that receives the contact modules **160**. The contact assemblies **140** are arranged in the cavity **134** at a mating end for mating with the mating electrical connector **106**. For example, the contact assemblies **140** are arranged in the cavity **134** in rows and columns. The cavity **134** may form a receptacle that receives the mating end of the mating electrical connector **106**. The walls of the connector housing **132** may be chamfered and have a lead-in surfaces to guide mating of the mating electrical connector **72** in the cavity **134**. The connector housing **132** may have guide features to properly position the mating electrical connector **72** within the cavity **134**. In an exemplary embodiment, the connector housing **132** includes a plurality of module slots **136** in the cavity **134**. Each module slot **136** receives one of the contact modules **160**. The contact modules **160** may be latchably coupled to the connector housing **132** in the corresponding module slots **136**.

[0040] With additional reference to FIG. 7, which is a front perspective view of one of the contact modules **160** in accordance with an exemplary embodiment, and FIG. 8, which is an enlarged view of the mating end of the contact module **160**, the contact module **160** includes a structure holding the contact assemblies **140** and the cables **150**. For example, the contact module **160** includes a body **162** holding the contact assemblies **140** and the cables **150**. The cables **150** extend from a cable end of the contact module **160**. The contact assemblies **140** extend from a mating end of the contact module **160**.

[0041] The body **162** may be generally rectangular having atop, a bottom, a front, a rear, a first side and a second side. The sides may be generally planar and parallel to each other configured to be stacked in the module stack side-by-side with the other contact modules **160**. In an exemplary embodiment, the body **162** holds a ground shield **164**. The body **162** includes latches **166**, such as at the top and the bottom of the body **162**, used to secure the contact module **160** in the connector housing **132** (FIG. 5). The contact module **160** may include a strain relief **168**, such as at the cable end, configured to hold the ends of the cables **150** and provide strain relief for the cables **150**. The strain relief **168** may be an overmold overmolded over the ends of the cables **150**.

[0042] Each contact assembly **140** includes at least one signal contact **142**, a contact holder **141** holding the at least one signal contact **142**, and a shield **144** coupled to the contact holder **141** and providing electrical shielding for the at least one signal contact **142**. Optionally, multiple contact assemblies **140** may be integrated into a common structure. For example, the contact holders **141** of each of the contact assemblies **140** may be a unitary structure, such as being a single molded piece holding all of the signal contacts **142** and the shields **144**.

[0043] In the illustrated embodiment, each contact assembly **140** includes a pair of the signal contacts **142**, which define a differential pair. In various embodiments, the signal contacts **142** may be deflectable spring beams. However, in alternative embodiments, the signal contacts **142** may be pin contacts, socket contacts, or other types of contacts. The signal contacts **142** may be stamped and formed contacts. Each signal contact **142** is configured to be electrically connected to a corresponding wire or conductor of the cable **150**. In an exemplary embodiment, the cable **150** is a twin-axial cable having a first conductor and a second conductor terminated to corresponding signal contacts **142**.

[0044] The contact holder **141** is manufactured from a dielectric material, such as a plastic material. The contact holder **141** may be a molded part. The contact holder **141** includes a contact channel **143** for each signal contact **142**. In the illustrated embodiment, the contact holder **141** has two contact channels **143** for the pair of signal contacts **142**. In various embodiments, the contact holder **141** is preformed and the signal contacts **142** are loaded into the contact holder **141**. In other various embodiments, the contact holder **141** is formed in situ around the signal contacts **142**, such

as being overmolded over the signal contacts **142**. The cable **150** may extend into an end of the contact holder **141** to terminate to the signal contacts **142**. The contact holder **141** may provide strain relief for the end of the cable **150**. In other embodiments, a single contact holder may be provided for the contact module **160** configured to hold each of the pairs of the signal contacts **142**, such as four pairs of the signal contacts **142** in a single molded part.

[0045] The shield **144** may be a stamped and formed shield. The shield **144** is configured to be coupled to the contact holder **141**. Optionally, shields may be provided at both sides of the contact module **160**. The shield **144** may provide shielding on multiple sides of the signal contacts **142**. In an exemplary embodiment, the front end of the shield **144** is C-shaped providing electrical shielding on three sides of the pair of signal contacts **142**. The shield **144** may have other shapes in alternative embodiments. The shield **144** may have deflectable spring beams configured to be mated to a shield structure of the mating connector assembly **104**. The shield **144** may be configured to be electrically connected to the cable shield of the cable.

[0046] With reference back to FIGS. 5 and 6, the connector holder **200** is used to hold the cable connector(s) **130** and the connector identification module **402** relative to the front panel **114**. The connector holder **200** is configured to be coupled to the cartridge **110**, via the mounting assembly **300**, to position each cable connector **130** and the connector identification module **402** relative to the cartridge **110**. In an exemplary embodiment, the cable connector **130** is movable relative to the connector holder **200** to position the cable connector **130** during mating with the mating electrical connector **72** (FIG. 4). For example, the connector holder **200**, and thus the cable connector **130** and the connector identification module **402**, may be movable along the mating axis (along the Z-axis) to accommodate overtravel of the mating connector assembly **70** during mating. In various embodiments, the connector holder **200**, and thus the cable connector **130** and the connector identification module **402**, may have a limited amount of floating movement relative to the front panel **114** to accommodate misalignment of the cable connector **130** relative to the mating electrical connector **72**. For example, the mounting assembly **300** allows movement of the connector holder **200** relative to the front panel **114** along the X-axis and/or along the Y-axis to accommodate misalignment of the cable connector **130** relative to the mating electrical connector **72**. The mounting assembly **300** may limit or confine the amount of floating movement in the lateral floating directions X/Y. For example, the mounting assembly **300** may limit floating movement, such as to approximately 5.0 mm for mating tolerance. In various embodiments, the mounting assembly **300** may limit floating movement to approximately 3.0 mm for mating tolerance.

[0047] The connector holder **200** includes a frame **202** defining a connector chamber **204**. The cable connector(s) **130** are received in the connector chamber **204**. The frame **202** supports the cable connector **130** in the connector chamber **204**. The connector identification module **402** may be coupled to the frame **202**. In various embodiments, the connector identification module **402** is mounted directly to the frame **202**. The connector identification module **402** may be located in the connector chamber **204**. In other various embodiments, the connector identification module **402** may be mounted to an exterior of the frame **202**, such as to a side of the frame **202**. The connector identification module **402** may be coupled to the frame **202** in alignment with the cable connectors **130**, such as in line with (for example, coplanar) the cable connectors **130**. The connector identification module **402** may be arranged side-by-side with the cable connector **130**.

[0048] The frame **202** extends between a front **206** and a rear **208** of the connector holder **200**. The mating ends of each cable connector **130** is provided at or forward of the front **206** of the connector holder **200** for mating with the mating electrical connector **72**. The mating end of the connector identification module **402** is provided at or forward of the front **206** of the connector holder **200** for mating with the mating identification module **80**. The cables **150** extends from the rear **208** of the connector holder **200**. The guide posts **122** extend forward from the front **206**, such as above the cable connector **130**. Other locations are possible in alternative embodiments, such as the bottom for one or both sides of the frame **202**.

[0049] In an exemplary embodiment, the frame **202** includes an upper frame member **210**, a lower frame member **212**, a first side frame member **214**, and a second side frame member **216**. The connector chamber **204** is defined between the upper frame member **210** and the lower frame member **212**. The connector chamber **204** is defined between the first and second side frame members **214**, **216**. In an exemplary embodiment, the connector identification module **402** is supported by one of the frame members, such as the first side frame member **214**. For example, the connector identification module **402** may be directly coupled to the first side frame member **214**, such as to the interior surface or the exterior surface of the first side frame member **214**.

[0050] The upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and/or the second side frame member **216** may be plates or blocks. For example, the upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and/or the second side frame member **216** may be stamped and formed plates. The upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and/or the second side frame member **216** may be die cast or molded blocks. In an exemplary embodiment, the upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and/or the second side frame member **216** may be discrete pieces secured together using fasteners, clips, latches, welding or other securing means. In other various embodiments, the upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and/or the second side frame member **216** may be integral with each other. The upper frame member **210**, the lower frame member **212**, the first side frame member **214**, and the second side frame member **216** form a brick **220** used to hold the cable connectors **130** and the connector identification module **402**. The brick **220** is a rectangular structure that surrounds the connector chamber **204**. The brick **220** is configured to be coupled to the cartridge **110** to hold the cable connectors **130** and the connector identification module **402** relative to the cartridge **110**.

[0051] In an exemplary embodiment, the frame **202** includes mounting features **224** configured to be coupled to the mounting assembly **300**. In the illustrated embodiment, the mounting features **224** include openings **226** in the frame members, such as in the upper frame member **210** and/or the lower frame member **212**. In the illustrated embodiment, four openings **226** are provided near the four corners of the brick **220**. In an exemplary embodiment, the mounting features **224** include keying features **228** for keyed mating with the mounting assembly **300**. For example, the keying features **228** may be ribs, tabs or other types of protrusions extending into the openings **226**. In other embodiments, the keying features **228** may be grooves, slots or other types of openings.

[0052] In an exemplary embodiment, the mounting assembly **300** is operably coupled between the connector holder **200** and the front panel **114**. The mounting assembly **300** allows the connector holder **200**, and thus the cable connector(s) **130** and the connector identification module **402** to move relative to the front panel **114**. For example, the mounting assembly **300** allows the connector holder **200** to move along the mating axis (Z-direction), such as to accommodate overtravel of the mating connector assembly **70** during mating. The mounting assembly **300** allows the connector holder **200**, and thus the cable connector **130** and the connector identification module **402**, to move in one or more lateral floating directions perpendicular to the mating direction, such as in a horizontal floating direction (X-axis) and/or a vertical floating direction (Y-axis).

[0053] In an exemplary embodiment, the mounting assembly **300** includes one or more locating assemblies **302** configured to interface with the connector holder **200** at the corresponding mounting features **224**. Each locating assembly **302** of the mounting assembly **300** includes a shaft **310**, a biasing member **330** operably coupled between the connector holder **200** and the shaft **310**, and a support element **350** configured to be operably coupled between the shaft **310** and the front panel **114** of the cartridge **110** to support the shaft **310** relative to the front panel **114**.

[0054] The support element **350** may be a front support configured to be coupled to the front panel **114** of the cartridge **110**. The support element **350** is coupled to the shaft(s) **310**, such as using fasteners **352**, clips, latches, welding, or other securing means. In an exemplary embodiment, the

support element **350** includes an inner surface **354** configured to be coupled to the front surface of the panel **114**. In an exemplary embodiment, the support element **350** includes an elongated plate, such as extending along the top and/or bottom and/or sides of the opening **120**. The support element **350** may be planar. The support element **350** may be coupled to multiple shafts **310**. The support element **350** may include notches **356** that receive the guide posts **122**. The support element **350** may have other shapes in alternative embodiments. For example, the support element **350** may be circular and configured to mate to a single shaft **310**.

[0055] The shaft **310** is configured to be coupled to the support element **350**. For example, the shaft **310** is configured to extend through a corresponding locating opening **121** in the front panel **114** to interface with the support element **350**. The front panel **114** may be captured between the support element **350** and the shaft **310**. The locating opening **121** may be at or near the port **120** in the front panel **114**, such as being open to the port **120**. In an exemplary embodiment, the shaft **310** is undersized relative to the locating opening **121** to allow a limited amount of floating movement in the locating opening **121** in a float direction (X-direction and/or Y-direction) perpendicular to the mating direction (Z direction) to allow the connector holder **200** and the cable connector **130** to move relative to the panel **114**. In an exemplary embodiment, the biasing member **330** forward biases the connector holder **200** and the cable connector **130**, such as to press the connector holder **200** toward the rear side of the panel **114**. The biasing member **330** is compressible in a rearward compression direction parallel to the mating direction (for example, along the Z axis) to allow the connector holder **200** and the cable connector **130** to move rearward relative to the panel **114**. As such, the mounting assembly **300** allows floating movement of the connector holder **200**, and thus the cable connector(s) **130**, in X/Y/Z directions to align the cable connectors **130** with the mating connectors **72** (FIG. **4**) during the mating process.

[0056] In an exemplary embodiment, the locating openings **121** are used to control floating movement of the connector assembly **102**, such as in the lateral floating direction(s) (for example, X direction and/or Y direction). In an exemplary embodiment, each locating opening **121** is oversized relative to the shaft **310**. For example, clearance gaps may be provided around the shaft **310** to move or float laterally within the locating opening **121**. In an exemplary embodiment, the locating opening **121** is horizontally oversized relative to the shaft **310** allowing the shaft **310** to move side-to-side within the locating opening **121** to allow a limited amount of lateral floating movement of the connector holder **200**, and thus the cable connector(s) **130** and the connector identification module **402**, relative to the front panel **114**. As such, the connector housing **132** is movable in the X-direction (for example, side-to-side) relative to the panel **114**. The amount of lateral movement may be limited to a confined amount by the edges of the panel defining the locating opening **121**. In an exemplary embodiment, the locating opening **121** is vertically oversized relative to the shaft **310** allowing the shaft **310** to move up-and-down within the locating opening **121** to allow a limited amount of lateral floating movement of the connector holder **200**, and thus the cable connector(s) **130** and the connector identification module **402**, relative to the front panel **114**. As such, the connector housing **132** is movable in the Y-direction (for example, up and down) relative to the panel **114**. The amount of lateral movement may be limited to a confined amount by the edges of the panel **114** defining the locating opening **121**.

[0057] FIG. **9** is a cross sectional view of a portion of the communication system **100** showing the mating connector assembly **70** mated with the connector assembly **102** showing the connector identification module **402** in accordance with an exemplary embodiment. FIG. **10** is a cross sectional view of a portion of the communication system **100** showing the mating connector assembly **70** mated with the connector assembly **102** showing the connector identification module **402** in accordance with an exemplary embodiment.

[0058] The cable connector **130** and the connector identification module **402** extend from the front of the connector holder **200** for mating with the mating connector assembly **70**. For example, the cable connector **130** and the connector identification module **402** extend through the opening **120**

in the front panel **114** for direct mating with the mating connector **72** and the mating identification connector **82**.

[0059] In an exemplary embodiment, the connector identification module **402** includes the identification device **410** and identification contacts **440** operably coupled to the identification device **410**. The identification contacts **440** may be signal contacts, power contacts, ground contacts, or other types of contacts. The identification contacts **440** are configured to be mated with the mating identification connector **82** to electrically connect the connector identification module **402** and the mating identification module **80**.

[0060] In an exemplary embodiment, the identification device **410** includes an electronic device **412** that stores unique identification data. The electronic device **412** may be an integrated circuit, such as a chip. In an exemplary embodiment, the electronic device **412** is an EEPROM device, such as an EEPROM chip or component. The electronic device **412** is programmable to store the identification data, such as a part number, a program name, a slot ID number, and the like. The electronic device **412** may be a surface mount component, such as having solder tails, solder pads, solder balls, or other conductors or leads for connection to another component, such as a circuit board or directly to the identification contacts **440**.

[0061] In an exemplary embodiment, the identification device **410** includes a printed circuit board **420** hosting the electronic device **412** and the identification contacts **440**, and may include other components mounted to the printed circuit board **420**, such as capacitors, resistors, a processor, a memory module, or another component. The electronic device **412** is mounted to the printed circuit board **420**. For example, the electronic device **412** may be soldered to the printed circuit board **420**. In an exemplary embodiment, the printed circuit board **420** includes circuits, such as pads, traces, vias, and the like. The circuits may define the identification contacts **440**. In alternative embodiments, the identification contacts **440** may be separate contacts, such as stamped and formed contacts, terminated to the printed circuit board **420** and extending from the printed circuit board **420** for mating with the mating identification connector **82**. In the illustrated embodiment, the identification contacts **440** are pads of the printed circuit board **420** arranged at a card edge **422** of the printed circuit board **420**. The card edge **422** is provided at a mating end of the identification device **410** for mating with the mating identification connector **82**. For example, the card edge **422** may be plugged into the card slot **84** of the mating identification connector **82**.

[0062] In various embodiments, the identification device **410** includes a device shell (not shown) holding the printed circuit board **420** and the electronic device **412**. The device shell surrounds the printed circuit board **420** and the electronic device **412** to protect the components. The device shell may be an overmold body overmolded over the printed circuit board **420** and the electronic device **412**. The device shell may be a composite plastic material. The card edge **422** may protrude from the device shell for mating with the mating identification connector **82**.

[0063] In an exemplary embodiment, the connector identification module **402** is received in the connector chamber **204** of the frame **202** of the connector holder **200**. The connector identification module **402** may be stacked adjacent to the cable connector **130**. In an exemplary embodiment, the connector identification module **402** includes locating features **428** for locating the connector identification module **402** in the connector chamber **204**. For example, the locating features **428** may be tabs or other protrusions received in a groove formed in the frame **202**, such as in the top and bottom frame members **210**, **212**. Other types of locating features may be used in alternative embodiments. The connector identification module **402** may be coupled to the first side frame member **214**. For example, the connector identification module **402** is coupled to an interior surface of the first side frame member **214**. The printed circuit board **420** and/or the device shell may be secured to the first side frame member **214**, such as using latches, clips, fasteners, and the like.

[0064] FIG. **11** is a perspective view of the connector identification module **402** in accordance with an exemplary embodiment. In an exemplary embodiment, the connector identification module **402** includes the identification device **410** and a device shell **430** holding the identification device **410**.

[0065] In an exemplary embodiment, the identification device **410** includes the printed circuit board **420** hosting the electronic device **412** and the identification contacts **440**, and may include other components mounted to the printed circuit board **420**, such as capacitors, resistors, a processor, a memory module, or another component. The electronic device **412** stores unique identification data. The electronic device **412** may be an integrated circuit, such as a chip. In an exemplary embodiment, the electronic device **412** is an EEPROM device, such as an EEPROM chip or component. The electronic device **412** is programmable to store the identification data, such as a part number, a program name, a slot ID number, and the like. The electronic device **412** may be a surface mount component, such as having solder tails, solder pads, solder balls, or other conductors or leads for connection to another component, such as a circuit board or directly to the identification contacts **440**.

[0066] The electronic device **412** is mounted to the printed circuit board **420**. For example, the electronic device **412** may be soldered to the printed circuit board **420**. In an exemplary embodiment, the printed circuit board **420** includes circuits, such as pads, traces, vias, and the like. The circuits may define the identification contacts **440**. In alternative embodiments, the identification contacts **440** may be separate contacts, such as stamped and formed contacts, terminated to the printed circuit board **420** and extending from the printed circuit board **420** for mating with the mating identification connector **82**. In the illustrated embodiment, the identification contacts **440** are pads of the printed circuit board **420** arranged at the card edge **422** of the printed circuit board **420**. The card edge **422** is provided at a mating end of the identification device **410** for mating with the mating identification connector **82**.

[0067] In an exemplary embodiment, the device shell **430** holds the printed circuit board **420** and the electronic device **412**. The device shell **430** surrounds the printed circuit board **420** and the electronic device **412** to protect the components. The device shell **430** may be an overmold body overmolded over the printed circuit board **420** and the electronic device **412**. The card edge **422** protrudes from the device shell **430** for mating with the mating identification connector **82**.

[0068] FIG. **12** is a top view of a portion of the communication system **100** in accordance with an exemplary embodiment showing the mating connector assembly **70** mated with the connector assembly **102** showing the connector identification module **402** shown in FIG. **11**. FIG. **13** is a side view of a portion of the communication system **100** in accordance with an exemplary embodiment showing the mating connector assembly **70** mated with the connector assembly **102** showing the connector identification module **402** shown in FIG. **11**. FIG. **14** is a rear perspective view of a portion of the communication system **100** in accordance with an exemplary embodiment showing the mating connector assembly **70** mated with the connector assembly **102** showing the connector identification module **402** shown in FIG. **11**.

[0069] The cable connector **130** and the connector identification module **402** extend from the front of the connector holder **200** for mating with the mating connector assembly **70**. For example, the cable connector **130** and the connector identification module **402** extend through the opening **120** in the front panel **114** for direct mating with the mating connector **72** and the mating identification connector **82**.

[0070] In an exemplary embodiment, the connector identification module **402** is coupled to the frame **202** of the connector holder **200**. The connector identification module **402** is arranged in line with the cable connectors **130**, such as horizontally aligned with the cable connectors **130**. The connector identification module **402** is coupled to the first side frame member **214**. For example, the connector identification module **402** is coupled to an exterior surface of the first side frame member **214**. The printed circuit board **420** and/or the device shell **430** may be secured to the first side frame member **214**, such as using latches, clips, fasteners, and the like.

[0071] It is to be understood that the above description is intended to be illustrative, and not restrictive. For example, the above-described embodiments (and/or aspects thereof) may be used in combination with each other. In addition, many modifications may be made to adapt a particular

situation or material to the teachings of the invention without departing from its scope. Dimensions, types of materials, orientations of the various components, and the number and positions of the various components described herein are intended to define parameters of certain embodiments, and are by no means limiting and are merely exemplary embodiments. Many other embodiments and modifications within the spirit and scope of the claims will be apparent to those of skill in the art upon reviewing the above description. The scope of the invention should, therefore, be determined with reference to the appended claims, along with the full scope of equivalents to which such claims are entitled. In the appended claims, the terms “including” and “in which” are used as the plain-English equivalents of the respective terms “comprising” and “wherein.” Moreover, in the following claims, the terms “first,” “second,” and “third,” etc. are used merely as labels, and are not intended to impose numerical requirements on their objects. Further, the limitations of the following claims are not written in means-plus-function format and are not intended to be interpreted based on 35 U.S.C. § 112(f), unless and until such claim limitations expressly use the phrase “means for” followed by a statement of function void of further structure.

Claims

1. A connector assembly for mating with a mating connector assembly, the connector assembly comprising: a connector holder having a frame defining a connector chamber, the frame extending between a front and a rear of the connector holder, the frame including an upper frame member, a lower frame member, and first and second side frame members extending between the upper and lower frame members to define the connector chamber, the connector holder configured to be mounted to a panel; a cable connector received in the connector chamber, the cable connector including a connector housing holding contact assemblies, each contact assembly including a signal contact and a cable terminated to the signal contact, the signal contact configured to be mated with a mating signal contact of a mating connector of the mating connector assembly, the cable connector having a mating end extending forward from the front of the frame and configured to pass through an opening in the panel for mating with the mating connector in a mating direction; and a connector identification module held by the frame and including a mating end configured to be mated with a mating identification module of the mating connector assembly, the connector identification module including identification contacts and an identification device, the identification contacts arranged at the mating end and configured to be mated with the mating identification module, the identification device having unique identification data stored on the identification device, the identification device being operably coupled to the identification contacts to transmit the unique identification data to the mating identification module.
2. The connector assembly of claim 1, wherein the connector identification module is coupled to the first side frame member.
3. The connector assembly of claim 1, wherein the connector identification module is aligned in plane with the cable connector.
4. The connector assembly of claim 1, wherein the connector identification module includes a printed circuit board, the identification module being mounted to the printed circuit board.
5. The connector assembly of claim 1, wherein the connector identification module includes a printed circuit board having a card edge, the identification contacts being circuits of the printed circuit board at the card edge, the card edge configured to be plugged into a card slot of the mating identification module.
6. The connector assembly of claim 1, wherein the connector identification module includes a module housing holding the identification device, the module housing being mounted to the first side frame member to position the connector identification module adjacent the cable connector to pass through the panel for mating with the mating identification module.
7. The connector assembly of claim 1, wherein the connector identification module is mounted to

an exterior surface of the frame.

8. The connector assembly of claim 1, wherein the connector identification module is received in the connector chamber.

9. The connector assembly of claim 1, wherein the identification device is an EEPROM device.

10. The connector assembly of claim 1, wherein the connector identification module includes shields providing shielding for the identification contacts.

11. The connector assembly of claim 1, further comprising a mounting assembly for mounting the connector holder to the panel, the mounting assembly including a shaft, a biasing member operably coupled between the connector holder and the shaft, and a support element configured to be operably coupled between the shaft and the panel to support the shaft relative to the panel, the shaft configured to be received in a locating opening, wherein the shaft is undersized relative to the locating opening to allow a limited amount of floating movement in the locating opening in a float direction perpendicular to the mating direction to allow the connector holder and the cable connector to move relative to the panel, the biasing member forward biasing the cable connector and being compressible in a rearward compression direction parallel to the mating direction to allow the connector holder and the cable connector to move rearward relative to the panel.

12. The connector assembly of claim 1, further comprising at least one guide pin extending forward of the frame to guide mating with the mating connector.

13. A connector assembly for mating with a mating connector assembly, the connector assembly comprising: a connector holder having a frame defining a connector chamber, the frame extending between a front and a rear of the connector holder, the frame including an upper frame member, a lower frame member, and first and second side frame members extending between the upper and lower frame members to define the connector chamber, the connector holder configured to be mounted to a panel; a cable connector received in the connector chamber, the cable connector including a connector housing holding contact assemblies, each contact assembly including a signal contact and a cable terminated to the signal contact, the signal contact configured to be mated with a mating signal contact of a mating connector of the mating connector assembly, the cable connector having a mating end extending forward from the front of the frame and configured to pass through an opening in the panel for mating with the mating connector in a mating direction; and a connector identification module held by the frame and including a mating end configured to be mated with a mating identification module of the mating connector assembly, the connector identification module including a printed circuit board having a card edge at the mating end, the connector identification module including identification contacts arranged at the card edge, the connector identification module including an identification device mounted to the printed circuit board, the identification device having unique identification data stored on the identification device, the identification device being operably coupled to the identification contacts, wherein the card edge is configured to be plugged into a card slot of the mating identification module when the connector assembly is mated with the mating connector assembly to transmit the unique identification data to the mating identification module.

14. The connector assembly of claim 13, wherein the connector identification module is coupled to the first side frame member and aligned in plane with the cable connector.

15. The connector assembly of claim 13, wherein the connector identification module includes a module housing holding the identification device, the module housing being mounted to the first side frame member to position the connector identification module adjacent the cable connector to pass through the panel for mating with the mating identification module.

16. The connector assembly of claim 13, wherein the connector identification module is received in the connector chamber.

17. The connector assembly of claim 13, wherein the identification device is an EEPROM device.

18. A communication system comprising: a cartridge having panels forming a chamber, the panels including a front panel, the cartridge including ports in the front panel at a front of the cartridge;

connector assemblies received in the ports for mating with mating connector assemblies, each connector assembly including a connector holder, a cable connector held by the connector holder, and a connector identification module for identifying the cable connector within the communication system, each connector holder having a frame defining a connector chamber, the frame extending between a front and a rear of the connector holder, the frame including an upper frame member, a lower frame member, and first and second side frame members extending between the upper and lower frame members to define the connector chamber, the connector holder being mounted to the cartridge at the corresponding port; each cable connector received in the corresponding connector chamber, each cable connector including a connector housing holding contact assemblies, each contact assembly including a signal contact and a cable terminated to the signal contact, the cable extending from the cable connector into the chamber of the cartridge, the signal contact configured to be mated with a mating signal contact of a mating connector of the corresponding mating connector assembly, the cable connector having a mating end extending forward from the front of the frame and configured to pass through the corresponding port in the front panel for mating with the mating connector in a mating direction; and each connector identification module held by the frame proximate to the cable connector, the connector identification module including a mating end configured to be mated with a mating identification module of the corresponding mating connector assembly, the connector identification module including identification contacts and an identification device, the identification contacts arranged at the mating end and configured to be mated with the mating identification module, the identification device having unique identification data stored on the identification device, the identification device being operably coupled to the identification contacts to transmit the unique identification data to the mating identification module.

19. The communication system of claim 18, wherein the connector identification module is coupled to the first side frame member and aligned in plane with the cable connector.

20. The communication system of claim 18, wherein the connector identification module includes a printed circuit board having a card edge, the identification contacts being circuits of the printed circuit board at the card edge, the card edge configured to be plugged into a card slot of the mating identification module.
